Supplementary Material
Pertaining to the manuscript:

《*A Trustworthy LLM-Enabled Workflow for Power-System Modeling and Security Assessment*》

by

Shanxiang Mao, Muyang Liu, Junru Chen

June, 10, 2026

Document Description

This supplementary document provides the benchmark prompts, experimental configurations, aggregated results, and failure-summary information supporting Section V of the main manuscript. Additional aggregated results and plotting-input tabl es are provided in the accompanying repository.

Contents

Section I. Benchmark Prompts

This section lists the 45 benchmark prompts used in the controlled 39-bus evaluation. The prompt suite is organized by three analytical domains, three task-complexity levels, and five instruction variants.

The analytical domains are:

- Domain A: steady-state analysis and OPF-related re-dispatch;
- Domain B: dynamic-stability assessment based on eigenvalue analysis;
- Domain C: security assessment and optimization involving OPF and N-1 contingency analysis.

The task-complexity levels are:

- L1: elementary tasks involving a single simulation function;
- L2: sequential tasks involving one parameter modification followed by analysis;
- L3: composite tasks involving cross-tool dependencies or intermediate state transitions.

The instruction variants represent different levels of procedural explicitness and implicit engineering assumptions. They are used to evaluate whether the workflow can maintain task-correct execution under different forms of engineering instruction.

Domain A:

- Level 1: Elementary Tasks
  - Var 1: "Run a power flow analysis on the IEEE 39-bus system."
  - Var 2: "Calculate the steady-state load flow solution for the 39-bus grid."
  - Var 3: "Please solve the power flow for the IEEE 39-node system."
  - Var 4: "Execute the power flow calculation for the 39-bus case."
  - Var 5: "I need the results of a standard load flow on the 39-bus model."
- Level 2: Sequential Tasks
  - Var 1: "Load the 39-bus system, set the active power of 'LOAD 15' to 150MW, and then run a power flow."
  - Var 2: "First modify 'LOAD 15' P_active to 150MW, then execute a load flow calculation."
  - Var 3: "Please set 'LOAD 15' P=150MW in the 39-bus system and calculate the new power flow."
  - Var 4: "I want to run a power flow after changing the 'LOAD 15' active power to 150MW on the 39-bus model."
  - Var 5: "After setting 'LOAD 15' to 150MW active power, solve the load flow for the 39-bus system."
- Level 3: Composite Tasks

    - Var 1: "Set active power of 'Load 15' to 150 MW. Run a load flow to update the state, then execute an Optimal Power Flow (OPF) to re-dispatch generation."
    - Var 2: "Simulate a load surge event on 'Load 15' (150 MW). Assess the system state and perform a generation cost optimization."
    - Var 3: "The system is facing a 150 MW load surge at Bus 15. Restore the grid to an economically optimal operating point."
    - Var 4: "Re-dispatch the generation to minimize cost accounting for the load increase to 150 MW at 'Load 15'."
    - Var 5: "Handle the load spike at 'Load 15' (150 MW) and ensure the operation is cost-efficient."

Domain B:

- Level 1: Elementary Tasks
    - Var 1: "Perform an eigenvalue analysis on the IEEE 39-bus system."
    - Var 2: "Calculate the system eigenvalues for the 39-bus grid."
    - Var 3: "Please run the small-signal stability analysis for the 39-bus model."
    - Var 4: "Execute the eigenvalue calculation for the 39-bus case."
    - Var 5: "I need the eigenvalue results for the IEEE 39-node system."
- Level 2: Sequential Tasks
    - Var 1: "Load the 39-bus system, set the inertia constant (H) of 'GEN 5' to 4.0, and then run the eigenvalue analysis."
    - Var 2: "For the 39-bus case, first modify 'GEN 5' inertia (H) to 4.0, then

perform the small-signal stability calculation."

- Var 3: "Please set 'GEN 5' H=4.0 in the 39-bus system and calculate the new system eigenvalues."
- Var 4: "I want to run an eigenvalue analysis after changing the 'GEN 5' inertia to 4.0 on the 39-bus model."
- Var 5: "After setting 'GEN 5' inertia constant to 4.0, solve for the eigenvalues ofthe 39-bus system."

▪ Level 3: Composite Tasks

- Var 1: "Run a cost-minimization optimal power flow, and then perform an eigenvalue analysis on the resulting steady-state operating point."
- Var 2: "Establish an optimal economic dispatch using optimal power flow, then check the system's small-signal stability."
- Var 3: "Evaluate the dynamic stability margin of the system after economic dispatch."
- Var 4: "Determine if the minimum-cost generation schedule introduces any strictly unstable oscillation modes."
- Var 5: "Verify the dynamic security of the optimal cost state."

Domain C:

▪ Level 1: Elementary Tasks

- Var 1: "Run an AC cost-minimization OPF for the IEEE 39-bus system."
- Var 2: "Perform an OPF on the 39-bus grid, minimizing total generation cost."
- Var 3: "Please solve the AC OPF for the 39-bus model to minimize cost."
- Var 4: "Execute a standard AC OPF for the 39-bus case, minimizing cost."
- Var 5: "I need the results of a cost-minimization OPF on the 39-bus system."

▪ Level 2: Sequential Tasks

- Var 1: "Load the 39-bus system, set the Pmax of 'GEN 2' to 600MW, and then run a AC cost-minimization optimal power flow."
- Var 2: "For the 39-bus case, first modify the 'GEN 2' Pmax constraint to 600MW, then solve the AC cost-minimization optimal power flow."
- Var 3: "Please set 'GEN 2' active power upper limit to 600MW and run the AC-optimal power flow for minimal cost."
- Var 4: "I want to run a AC cost-minimization optimal power flow after changing 'GEN 2' Pmax to 600MW."
- Var 5: "Re-optimize the dispatch assuming 'GEN 2' is capped at 600MW."

▪ Level 3: Composite Tasks

- Var 1: "Run a cost-minimization optimal power flow. Explicitly save the result as the new system state, then run a full N-1 contingency analysis."
- Var 2: "Perform a secure-economic assessment: Optimize the cost first, then verify the N-1 security of that optimal state."
- Var 3: "Verify if the economically optimized dispatch is secure against N-1 contingencies."

- Var 4: "Assess the static security margins ofthe minimum-cost operating point against N-1 faults."
- Var 5: "Find the cheapest operating point and stress-test it against all N-1 faults."

Section II. System Prompt

The following system prompt was used for workflow planning in the full-workflow configuration.

Role and Purpose: You are an expert Power System Agent. Your primary purpose is to bridge the semantic gap between high-level engineering intents and executable power system simulation workflows.

Primary Function: You must analyze the user's request, infer implicit dependencies, and formulate a logical, physically sensible sequence of operations (a workflow plan) to achieve the specified engineering goals.

Execution Constraints:

1. Every step in your planned sequence must strictly adhere to the designated JSON format. The task_type field must be explicitly declared at both the step level and within the task_details object.
2. You must infer and populate all necessary parameters for each task based on the user's input and the provided Tool Schema.
3. If the user's input lacks essential parameters required for execution, set the task status to clarification_needed. Otherwise, set it to completed.

Reasoning Guidelines:

- Translate vague user terminology (e.g., "assess impact", "optimize") into the most appropriate operational tool based strictly on the Schema descriptions.
- Ensure modifications (e.g., load changes, parameter adjustments) are scheduled strictly before the analytical tasks that depend on them.
- Engineering Best Practices:
    - *Verification-before-Analysis:* Typically, after modifying system parameters, schedule a steady-state verification (power_flow) to check for basic physical violations before proceeding to advanced analyses.
    - If the user's request is exploratory or cautious (e.g., "assess impact"), include the intermediate power_flow. If the request is highly urgent or strictly goal-oriented (e.g., "restore operation", "minimize cost immediately"), you may prioritize efficiency by skipping the intermediate check and directly scheduling the optimization tool.
- State-Dependency Tracking: If an analysis depends on a newly optimized or modified state (e.g., "Run N-1 on the optimized state"), explicitly schedule a state-saving step (e.g., save_workpoint) before the subsequent analysis.

Examples (Few-Shot):

- **User:** "Run a power flow analysis." **Plan:** [power_flow]
- **User:** "Set Load 15 to 100MW and run power flow." **Plan:** [modify_load, power_flow]

(*Note: Complex multi-step reasoning must be inferred autonomously based on the guidelines above.*)

Section III. Experimental Configurations

This section lists the evaluated configurations and the repeated-trial protocol used in the controlled 39-bus benchmark. Each configuration uses the same 45-prompt suite. Each prompt is repeated 30 times, resulting in 1350 runs per configuration and 6750 runs in total.

3.1 Evaluated configurations

Configuration	Description	LLM backbone
Full workflow	Complete workflow with schema guidance and deterministic orchestration	Qwen2.5-32B-Instruct
Schema guidance removed	Natural-language schema/workflow guidance is removed while the benchmark protocol is unchanged	Qwen2.5-32B-Instruct
Control layer removed	Deterministic workflow control is bypassed or weakened	Qwen2.5-32B-Instruct
Qwen2.5-7B	Model-sensitivity configuration with a smaller instruction-tuned backbone	Qwen2.5-7B-Instruct
Llama-3.1-8B	Model-sensitivity configuration with an alternative instruction-tuned backbone	Meta-Llama-3.1-8B-Instruct

3.2 Repeated-trial protocol

For each configuration, the same 45-prompt benchmark suite is used. The prompts cover three analytical domains, three task-complexity levels, and five instruction variants. Each prompt is executed for 30 repeated trials under the corresponding configuration. The repeated-trial setting is used to reduce the influence of individual stochastic responses and to provide stable aggregate statistics for planning validity, task-correct execution, and the plan-to-execution gap.

The request-level generation settings and model-serving information are recorded in the run manifests. These include the evaluated configuration name, model backbone, repeated-trial setting, model path or model identifier, and generation parameters used by the serving backend.

3.3 Experimental record fields

Field	Description
prompt_id	Identifier of the benchmark prompt, including domain, complexity level, and instruction variant
domain	Analytical domain of the prompt
level	Task-complexity level: L1, L2, or L3
variant	Instruction variant: Var1–Var5
planning_success	Whether the generated workflow plan satisfies the planning-validity checks
execution_validity	Whether the executed workflow satisfies the intended task and postcondition checks
planning_fail_reason	Recorded reason for planning-stage failure, if applicable
execution_fail_reason	Recorded reason for execution-stage or postcondition failure, if applicable
final_status	Final workflow status of the run

The aggregated results reported in Section IV are computed from these prompt-level records. The repository provides the raw records and aggregation scripts used to produce the tables and plotting inputs reported in the main manuscript.

Section IV. Aggregated Results and Failure Summary

4.1 Full-workflow reliability

Task complexity	Planning validity (%)	Task-correct execution rate (%)	Plan-to-execution gap (%)
Elementary tasks	100.0	98.9	1.1
Sequential tasks	100.0	98.7	1.3
Composite tasks	97.4	94.7	2.7
Overall	99.1	97.4	1.7

4.2 Ablation and model-sensitivity results

Configuration	L1 execution (%)	L2 execution (%)	L3 execution (%)	Overall execution (%)	Overall gap (%)
Full workflow	98.9	98.7	94.7	97.4	1.7
Schema guidance removed	91.8	69.3	48.9	70.0	3.6
Control layer removed	44.7	90.4	68.2	67.8	22.3
Qwen2.5-7B	84.9	66.9	64.4	72.1	9.1
Llama-3.1-8B	86.9	60.2	44.4	63.9	0.3

4.3 Failure-summary statistics

Configuration	Planning-stage failures	Execution-stage failures	Main observed failure pattern
Full workflow	12	23	Missing intermediate operations or postcondition mismatches in a small number of composite tasks
Schema guidance removed	357	48	Missing required simulation operations during planning
Control layer removed	133	302	Execution-stage failures caused by incomplete parameters, state inconsistency, or failed solver-side execution

Qwen2.5-7B	254	123	Missing required operations and incomplete execution parameters
Llama-3.1-8B	483	5	Invalid or incomplete workflow plans before execution

A representative boundary case occurs in composite workflows that connect OPF and downstream security assessment. The workflow may correctly identify that an OPF calculation should be performed before N-1 contingency analysis, but fail to save or activate the optimized operating point before launching the downstream assessment. In this case, the individual solver calls may be valid, but the N-1 analysis is not guaranteed to be performed on the intended post-OPF system state. This illustrates why the benchmark evaluates task-correct execution rather than only syntactic plan validity or individual API executability.